

Branching

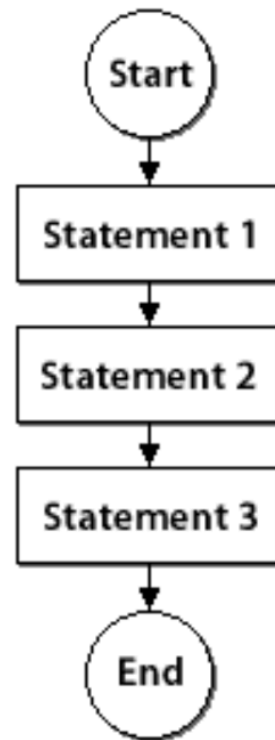
Introduction to computer programming

Management and Artificial Intelligence

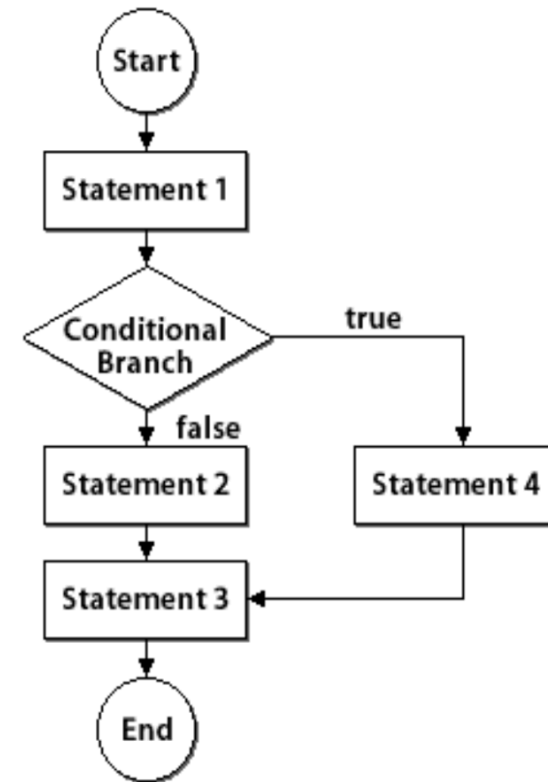


Branching

A program **without branching**



A program **with branching**



Branching in Python

Plain if

```
if <condition>:  
    statement  
    statement  
    ...
```

if-else

```
if <condition>:  
    statement  
    ...  
else:  
    statement  
    ...
```

if-elif-else

```
if <condition>:  
    statement  
    ...  
elif <condition>:  
    statement  
    ...  
else:  
    statement  
    ...
```

- `<condition>` yields `True` or `False`
- Evaluate statements in that block if `<condition>` is `True`
- Indentation is the key to delimit blocks
- Place `:` at the end of `<condition>`

Examples: if

```
In [16]: num = 3
if num > 0:
    print(num, "is a positive number.")
print("This is always printed. It's outside of the block.")
```

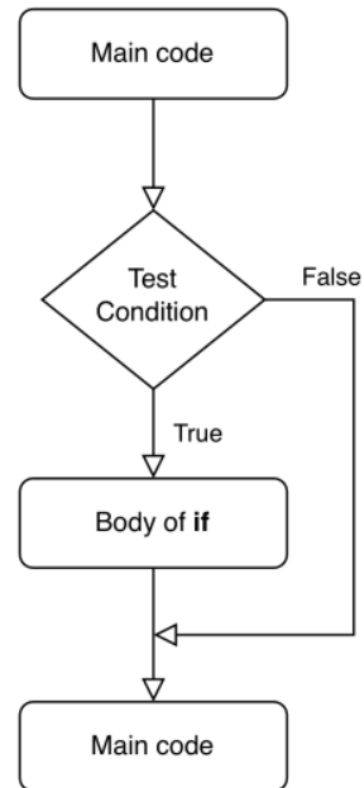
```
3 is a positive number.
This is always printed. It's outside of the block.
```

```
In [17]: num = -1
if num > 0:
    print(num, "is a positive number.")
print("This is always printed. It's outside of the block.")
```

```
This is always printed. It's outside of the block.
```

Flowchart of an **if** statement

Logical flowchart of an if statement:



Examples: if-else

```
In [18]: num = 1
if num >= 0:
    print("Number is positive or zero.")
else:
    print("Negative number.")
```

Number is positive or zero.

What happens with:

- num = 0
- num = -5

?

```
In [19]: num = 0
if num >= 0:
    print("Number is positive or zero.")
else:
    print("Negative number.")
```

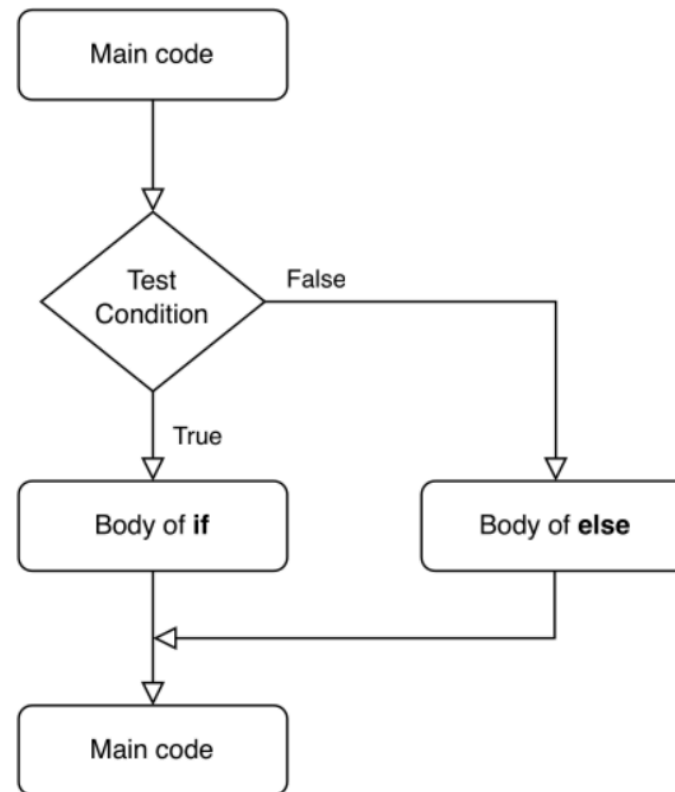
Number is positive or zero.

```
In [20]: num = -5
if num >= 0:
    print("Number is positive or zero.")
else:
    print("Negative number.")
```

Negative number.

Flowchart of an **if-else** statement

Logical flowchart of an if-else statement.:



Examples: if-elif-else

```
In [21]: num = 3.4
if num > 0:
    print("Number is positive.")
elif num == 0:
    print("Number is zero.")
else:
    print("Number is negative.")
```

Number is positive.

What happens with:

- num = 0
- num = -4.5

?

```
In [22]: num = 0
if num > 0:
    print("Number is positive.")
elif num == 0:
    print("Number is zero.")
else:
    print("Number is negative.")
```

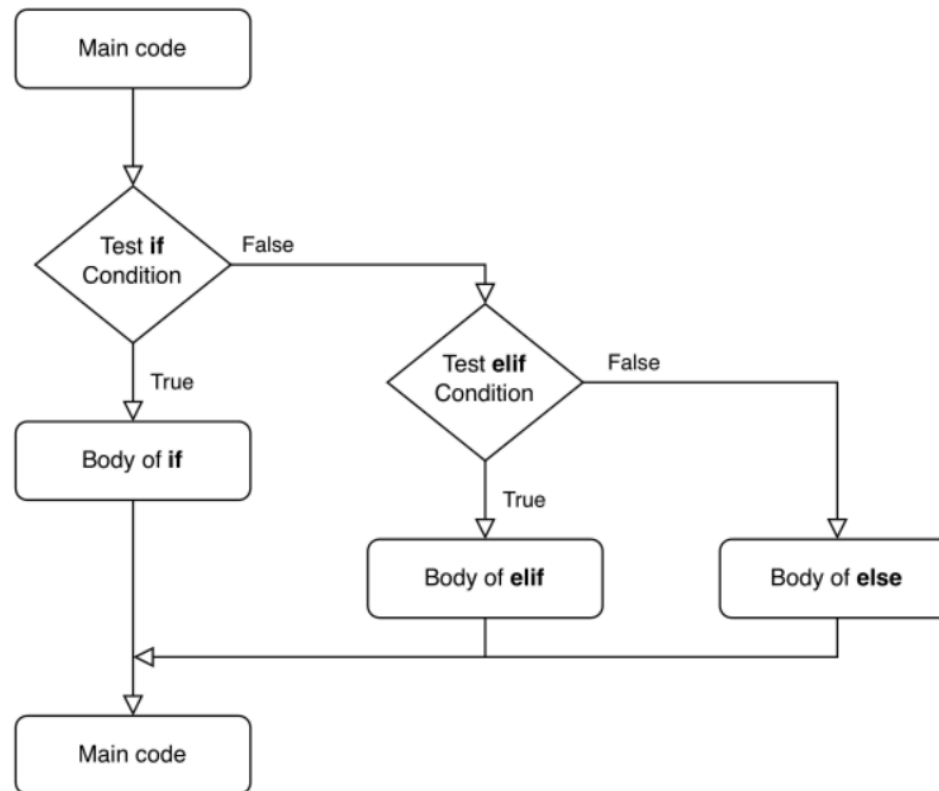
Number is zero.

```
In [23]: num = -4.5
if num > 0:
    print("Number is positive.")
elif num == 0:
    print("Number is zero.")
else:
    print("Number is negative.")
```

Number is negative.

Flowchart of an **if-elif-else** statement

Logical flowchart of an if-elif-else statement:



Nested Statements

Branching statements `if`, `else` and `elif` can be nested arbitrarily.

```
In [35]: num = 10
# or interactive:
# num = float(input()) # recall: input() returns a string
print("You entered: ", num)
if num >= 0:
    if num == 0:
        print("Number is zero")
    else:
        print("Number is positive")
else:
    print("Number is negative")
```

```
You entered: 10
Number is positive
```

Indentation

Recall: Indentation (tab or 4 spaces) delimits blocks of code in Python.

```
In [25]: x = float(input("Enter a value for x: "))
y = float(input("Enter a value for y: "))
print("x is", x)
print("y is", y)
if x == y:
    print("y and x are equal")
    if y != 0:
        print("Finally, x/y is", x / y)
elif x > y:
    print("y is smaller")
else:
    print("x is smaller")
```

```
x is 10.0
y is 20.0
x is smaller
```

Exercises

Exercise 1

Write a piece of code that checks whether a number is even. If so, print a message.

Exercise 1

Write a piece of code that checks whether a number is even. If so, print a message.

A possible solution follows:

```
In [27]: num = 42
if num % 2 == 0:
    print("Condition evaluates to True.")
    print("Number", num, "is even")
```

```
Condition evaluates to True.
Number 42 is even
```

Exercise 2

Write a piece of code that checks whether a number is even. Print a message about the outcome (odd or even).

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Write a piece of code that checks whether a number is even. Print a message about the outcome (odd or even).

A possible solution follows:

```
In [28]: num = 42
if num % 2 == 0:
    print("Condition evaluates to True.")
    print("Number", num, "is even")
else:
    print("Condition evaluates to False.")
    print("Number", num, "is odd")
```

```
Condition evaluates to True.
Number 42 is even
```

Exercise 3

Write a piece of code that checks whether a number is odd or even. Assign a proper value (e.g., 'odd' or 'even') to a variable, depending on the outcome of the condition. Print the outcome at the end.

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Write a piece of code that checks whether a number is odd or even. Assign a proper value (e.g., 'odd' or 'even') to a variable, depending on the outcome of the condition. Print the outcome at the end.

A possible solution follows:

```
In [29]: num = 42
if num % 2 == 0:
    parity = "even"
else:
    parity = "odd"
print("Number", num, "is", parity)
```

Number 42 is even

Exercise 4

Write a piece of code that checks whether a number is divisible by 2, 3, 5 and 7.

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Write a piece of code that checks whether a number is divisible by 2, 3, 5 and 7.

A possible solution follows:

```
In [30]: num = 42
if num % 2 == 0:
    print("Number ", num, "is divisible by 2.")
if num % 3 == 0:
    print("Number ", num, "is divisible by 3.")
if num % 5 == 0:
    print("Number ", num, "is divisible by 5.")
if num % 7 == 0:
    print("Number ", num, "is divisible by 7.")
```

Number 42 is divisible by 2.

Number 42 is divisible by 3.

Number 42 is divisible by 7.

Exercise 5

Write a piece of code that checks whether a number is odd or even. Print a message informing the user. If odd, check whether the number is divisible by 5. Print a message informing the user. If even, check whether the number is divisible by 3. Print a message informing the user.

Exercise 5

Write a piece of code that checks whether a number is odd or even. Print a message informing the user. If odd, check whether the number is divisible by 5. Print a message informing the user. If even, check whether the number is divisible by 3. Print a message informing the user.

A possible solution follows:

```
In [31]: num = 15
if num % 2 == 0:
    print("Number", num, "is even.")
    if num % 3 == 0:
        print("Number", num, "is also divisible by 3.")
    else:
        print("Number", num, "is not divisible by 3.")
else:
    print("Number", num, "is odd.")
    if num % 5 == 0:
        print("Number", num, "is also divisible by 5.")
    else:
        print("Number", num, "is not divisible by 5.")
```

Number 15 is odd.

Number 15 is also divisible by 5.

Exercise 6

Write a piece of code that asks the user to insert the dog's age in human's years and calculates a dog's age in dog's years. For the first two years, a dog year is equal to 10.5 human years. After that, each dog year equals 4 human years.

Exercise 6

Write a piece of code that asks the user to insert the dog's age in human's years and calculates a dog's age in dog's years. For the first two years, a dog year is equal to 10.5 human years. After that, each dog year equals 4 human years.

A possible solution follows:

```
In [36]: h_age = 19
# or interactive:
# h_age = int(input("Input a dog's age in human years: "))
print("Inserted age:", h_age)
if h_age <= 0:
    print("Age must be a positive number.")
else:
    if h_age <= 2:
        d_age = h_age * 10.5
    else:
        d_age = 10.5 + 10.5 + (h_age - 2) * 4
    print("The dog's age in dog's years is:", d_age)
```

Inserted age: 19

The dog's age in dog's years is: 89.0

